| NAME | GAYATHRI.P |
|---|---|
| REG NO | 192424324 |
| COURSE | Mobile Computing |
| CODE | ITA0329 |

# ASSIGNMENT – 1 -CO1

## TASK 1 – Design Alternative Frequency-Planning Strategies

Question: How can a fixed block of spectrum be shared across many cells so that every subscriber is served without intolerable co-channel interference, and in how many fundamentally different ways can this be done?

**Answer**

Three different strategies are considered. Each strategy shares the same 336 channels spatially, but controls interference and capacity in a different way.

**Strategy 1: Fixed Cluster Frequency Reuse**

Basic idea: Divide the cells into clusters. Each cluster contains N cells, and every cell in a cluster receives a different channel group. The same group is reused in another cluster far enough away to control co-channel interference. For N = 7, each cell receives 48 channels.

Design principle: Spatial reuse.

Access method: In an FDMA system, channels are separated by frequency. In TDMA, each carrier is divided into time slots. Control channels carry registration, paging and call setup; traffic channels carry voice/data.

Assumptions: Hexagonal geometry, approximately uniform traffic, fixed channel allocation and similar propagation conditions.

Iterative procedure / pseudocode:

1. Create a hexagonal grid for the 12 km² township.
2. Generate valid cluster sizes using $N = i^2 + ij + j^2$.
3. For each N, calculate channels per cell = 336/N.
4. Calculate reuse distance $D = R\sqrt{(3N)}$ and estimated SIR.
5. Reject cluster sizes that do not meet the selected SIR threshold.
6. Assign a different channel set to each cell in one cluster and repeat the pattern.
7. Calculate blocking and total supported traffic; select a feasible design.

Why investigate it? It is simple, predictable and easy to plan. Its weakness is that a fixed channel set can be underused in a quiet cell while another cell is congested.

**Strategy 2: Sectorised Frequency Reuse**

Basic idea: Divide a cell into three 120-degree sectors using directional antennas. Each sector mainly serves mobiles in its direction. Directional antennas reduce the number of strong co-channel interferers seen by a mobile.

Design principle: Spatial filtering.

Access method: FDMA/TDMA channel groups can be assigned to sectors. Control signalling supports access and handoff; traffic channels carry user calls. OFDMA can allocate groups of subcarriers to users, while CDMA separates users by codes.

Assumptions: Directional antennas are correctly aligned, antenna patterns suppress signals outside the intended sector, and sector traffic is reasonably balanced.

Pseudocode:

8. Create the hexagonal cell grid.

9. Divide each cell into three 120-degree sectors.

10. Assign sector-specific channel groups using the reuse plan.

11. Estimate the desired signal and interference from co-channel sectors.

12. Calculate SIR for users at different angles and distances.

13. Adjust antenna tilt, azimuth and power until the SIR target is met.

Why investigate it? Sectoring can improve SIR without increasing licensed spectrum. It adds antenna, installation and optimisation costs and may reduce trunking efficiency if channels are rigidly separated.

**Strategy 3: Dynamic Channel Assignment and Hierarchical Cell Splitting**

Basic idea: Channels are assigned according to current traffic and interference. A congested cell may borrow an idle channel from a neighbouring cell if the reuse and interference conditions are safe. For long-term growth, overloaded macro cells are split into smaller microcells.

Design principle: Trunking efficiency and hierarchical reuse.

Access method: The controller selects an available FDMA/TDMA channel or OFDMA resource block. Control channels coordinate requests, channel allocation and handoff.

Assumptions: The network has a controller or scheduler, neighbouring-cell load information is available, and borrowed channels pass an interference check.

Recursive / hierarchical procedure:

14. Measure traffic and blocking in every cell.

15. If a cell is below the overload threshold, keep its current size.

16. If it is overloaded, first search for a safe idle channel in a neighbour.

17. If borrowing is not enough, split the cell into four smaller cells of radius approximately R/2.

18. Recalculate channels, transmit power, interference and handoff load.

19. Repeat until demand is supported or the minimum practical cell size is reached.

Why investigate it? Dynamic allocation uses idle channels more efficiently during uneven demand. Cell splitting provides long-term capacity, but increases site cost, handoffs and planning complexity.

**Preprocessing and cost responsibility**

- Traffic map: estimate subscribers and busy-hour traffic by academic blocks, hostels, residential areas and stadium.
- Propagation survey: measure received signal strength, shadowing, building loss and interference.
- Hex-grid layout: map cell centres, candidate tower sites, roads and coverage boundaries.

The mobile operator normally bears radio-planning, survey, site acquisition, equipment and optimisation costs. The university or township authority may assist with access to land, building plans and permissions, depending on the agreement.

## TASK 2 – Identify the Governing Design Parameter

Question: For each strategy, which single quantity most strongly controls capacity and interference, and why should the analysis be built around it?

**Answer**

| **Strategy** | **Main governing parameter** | **Reason and growth response** |
|---|---|---|
| Fixed reuse | Cluster size N (or reuse ratio Q = D/R) | Larger N increases reuse distance and SIR but reduces channels per cell (336/N). As demand grows, use smaller cells or better interference control |

| | | |
|---|---|---|
| | | rather than reducing N below the SIR limit. |
| Sectoring | Number of effective co-channel interferers | Directional antennas reduce the number of strong interferers. As demand grows, sector capacity and antenna patterns must be checked. |
| Dynamic allocation / splitting | Cell offered load A and available channels | Borrowing depends on spare channels; splitting is triggered when offered load exceeds capacity. As subscribers grow, more cells or resources are required. |

The cluster size N is the main parameter for fixed reuse because it directly controls both channel count per cell and reuse distance. The offered traffic per cell is the main trigger for dynamic assignment and splitting.

## TASK 3 – Derive Capacity and Interference Functions Mathematically

Question: Starting from hexagonal geometry, how do coverage, capacity and interference depend on the design choices?

**Answer**

**1. Cluster-size relation**

For a hexagonal cellular layout, a cluster is reached by moving i cells in one direction and j cells after a 60° turn. The valid cluster size is:

$N = i^2 + ij + j^2$

Examples: (i,j) = (1,1) gives N = 3; (2,0) gives N = 4; (2,1) gives N = 7; (2,2) gives N = 12. Therefore N = 7 is a valid choice.

**2. Reuse distance and reuse ratio**

The distance between co-channel cell centres is $D = R\sqrt{(3N)}$. The reuse ratio is $Q = D/R = \sqrt{(3N)}$.

For N = 7: $Q = \sqrt{21} = 4.583$. If R = 0.25 km, $D = 0.25 \times 4.583 = 1.146$ km.

**3. Hexagonal cell area and number of cells**

The area of one regular hexagonal cell with cell radius R (centre to vertex) is $A_{cell} = (3\sqrt{3}/2)R^2 = 2.598R^2$.

For R = 0.25 km: $A_{cell} = 2.598 \times 0.25^2 = 0.1624$ km².

Required cells = township area / cell area = 12 / 0.1624 = 73.9. Therefore approximately 74 cells are required. A practical design adds boundary and overlap allowance.

**4. Channels per cell**

Channels per cell C = S/N = 336/7 = 48 channels.

**5. Erlang-B traffic capacity**

Erlang-B blocking probability is calculated recursively: B(0,A)=1 and $B(m,A) = [A \times B(m-1,A)]/[m + A \times B(m-1,A)]$. For 48 channels, solve B(48,A)=0.02. The offered traffic is approximately 38.39 Erlangs per cell.

Approximate total offered traffic capacity = 74 × 38.39 = 2,840.9 Erlangs. At 0.025 Erlang per subscriber, this corresponds to 2,840.9/0.025 ≈ 113,636 subscribers under the ideal equal-load assumption.

| **Subscribers** | **Traffic (Erlangs)** | **Approximate 74-cell capacity** | **Planning result** |
|---|---|---|---|
| 1,000 | 25 | 2,841 | Supported |
| 10,000 | 250 | 2,841 | Supported |
| 20,000 | 500 | 2,841 | Supported |
| 100,000 | 2,500 | 2,841 | Supported, but limited spare capacity |

This is a theoretical estimate. Real networks need reserve capacity for uneven traffic, control signalling, handoff, outages and interference.

**6. Co-channel SIR derivation**

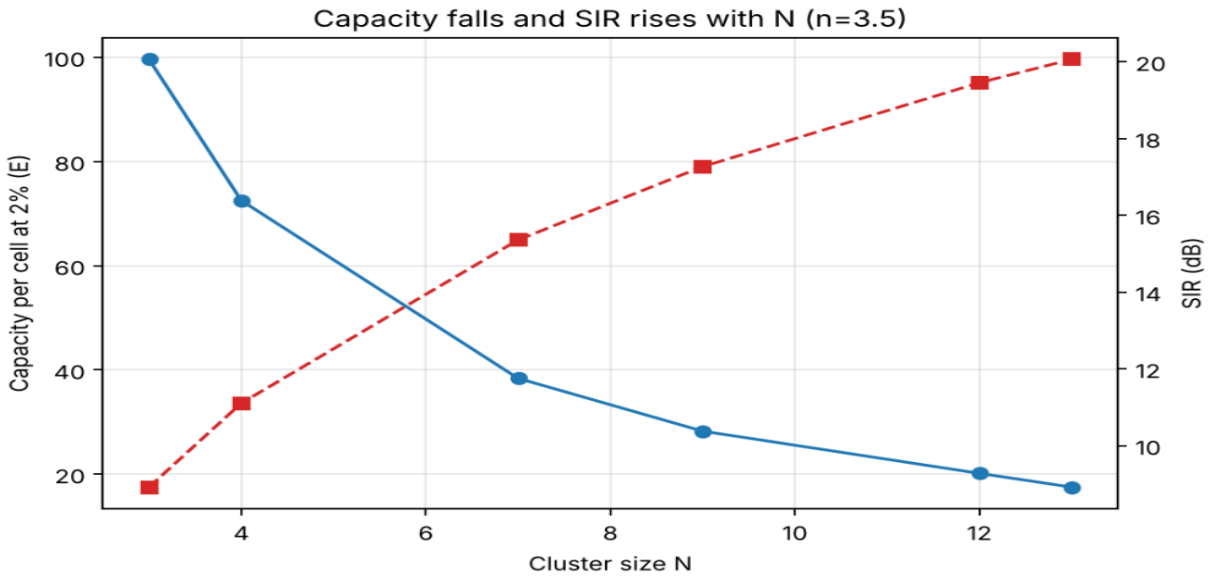
The received power from a transmitter is proportional to d^(−n), where d is distance and n is the path-loss exponent. For a mobile near the cell edge, desired distance is approximately R and the first-tier co-channel interferers are approximately D away.

With six equal-distance first-tier interferers: SIR ≈ (D/R)^n / 6 = (√(3N))^n / 6 = (3N)^(n/2)/6.

For N = 7 and n = 4: SIR = $21^2/6$ = 73.5. In dB, 10log10(73.5) = 18.66 dB, approximately 18.7 dB. This matches the assignment sanity-check.

For N = 7 and n = 2.5: SIR = 21^1.25/6 ≈ 11.5, or about 10.6 dB. This is below the selected 12 dB target, so open-campus areas need sectoring, power control, a larger reuse cluster, or a combination.

For ideal three-sector 120° sectoring, assume two dominant first-tier interferers instead of six. Then SIR ≈ (3N)^(n/2)/2. For N=7, n=2.5: SIR ≈ 34.5, or about 15.4 dB. This simplified model exceeds 12 dB. Actual antenna patterns and geometry must be simulated.




**7. Cost–capacity relationship**

Let C_survey be the one-time survey and planning cost, c_site the cost per site, and M the number of sites. Total initial planning/deployment cost can be represented as C_total = C_survey + M × c_site + C_optimisation. For the selected radius, M is approximately 74 before site-sharing adjustments.

A simple cost per offered Erlang is Cost/Erlang = C_total / (M × 38.39). A smaller radius increases M approximately in proportion to $1/R^2$, which increases site cost but improves spatial capacity. Fixed reuse has predictable cost; sectoring adds antenna cost; dynamic allocation adds controller/software and optimisation cost; cell splitting adds new sites and backhaul.

The cost model is symbolic because the assignment does not provide currency prices for surveys, towers, radios or backhaul. Actual monetary cost must be filled using local operator quotations.

## TASK 4 – Analyse Best, Average and Worst-Case Behaviour

Question: Where in the cell, and at what time of day, does each strategy perform best, typically and worst — and why?

**Answer**

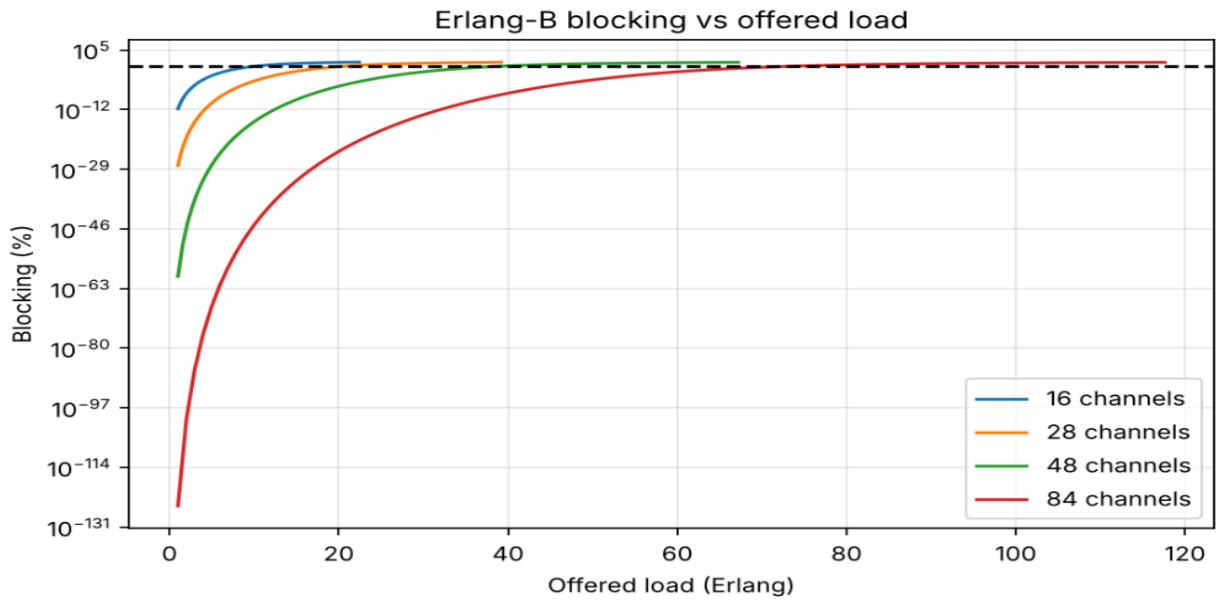
| Case | Mobile position / traffic | SIR and blocking explanation |
|---|---|---|
| Best | Near base station; off-peak traffic | Desired signal is strong and offered traffic is low. SIR is usually high and blocking is close to zero. |
| Average | Mid-cell; normal busy-hour traffic | Desired signal is weaker than near the base station. Interference and traffic are moderate; blocking depends on Erlang-B load. |
| Worst | Cell edge or corner; event surge | Desired signal is weak and multiple co-channel interferers may be visible. Offered traffic may exceed available channels, increasing blocking and outage. |

Illustrative SIR calculation for N=7 and n=4 using the six-interferer approximation: at the cell edge, desired distance is R and interferer distance is D=4.583R, giving SIR ≈ 18.66 dB. At a point closer to the base station, the desired distance is smaller; if the interferers remain approximately at D, the desired signal becomes stronger and SIR improves. Exact values depend on mobile angle and actual interferer distances.

At the cell corner, the mobile may be farther from its serving base station and closer to one or more neighbouring co-channel cells. The equal-distance approximation can therefore overestimate or underestimate SIR; a position-based calculation is needed.

Blocking example: for 48 channels, Erlang-B gives about 2% blocking at 38.39 Erlangs. At a load above 38.39 Erlangs, blocking exceeds the target. At low traffic, the same channel group has much lower blocking.

Shadowing can reduce the desired signal or increase an interfering signal. A larger path-loss exponent (n=4) attenuates distant interferers more strongly than n=2.5, generally improving reuse SIR. However, buildings can also create local coverage holes. Off-peak, average, busy-hour and event conditions must be evaluated separately.




## TASK 5 – Scaling and Sensitivity Analysis

Question: How do capacity, interference and the number of cells scale as demand and the environment change, and when does each strategy stop being practical?

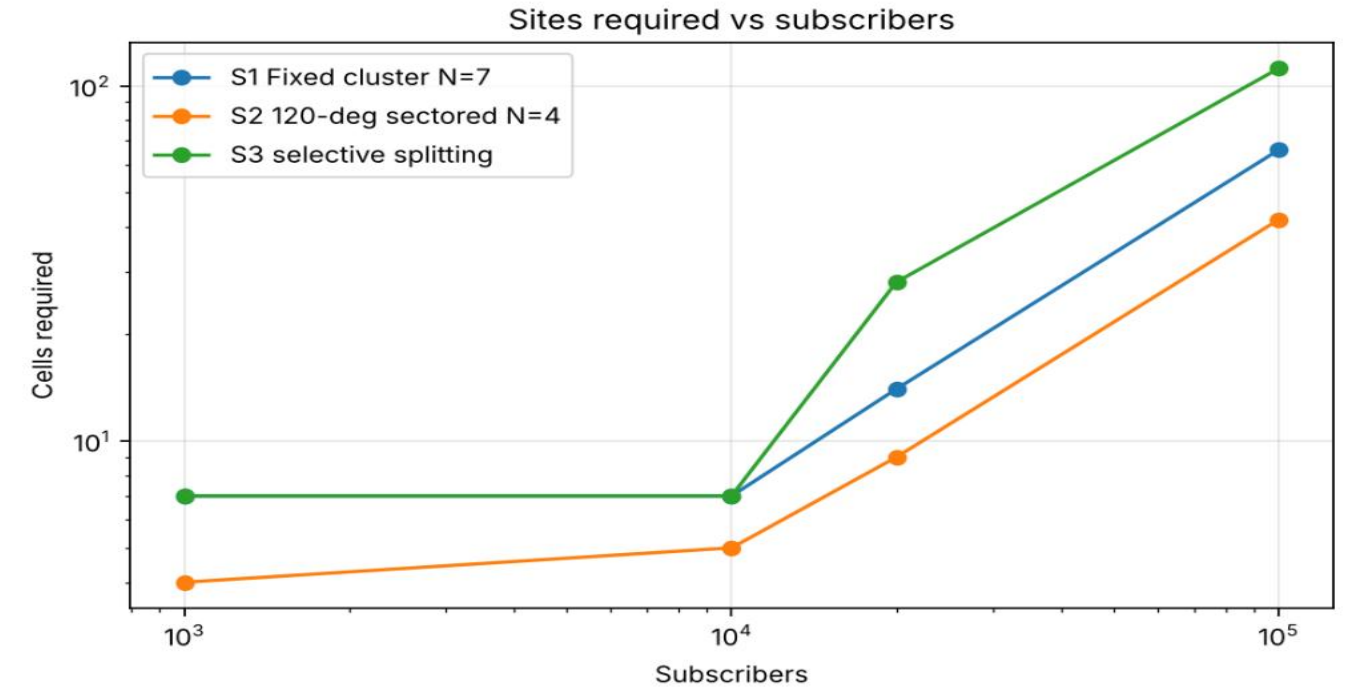
**Answer**

- Fixed reuse: channels per cell = S/N, so per-cell channel capacity is approximately proportional to 1/N. Reuse distance ratio Q = √(3N). Under the six-interferer model, SIR is proportional to N^(n/2).

- Coverage: $A_{cell} = 2.598R^2$, so number of cells $M \approx$ township area/$(2.598R^2)$, which is proportional to $1/R^2$.
- Cell splitting: reducing radius from R to R/2 makes each cell area one quarter; approximately four times as many cells are needed in the split region.
- Propagation: increasing n from 2.5 to 4 causes distant interferers to decay faster, improving co-channel SIR, but dense buildings may increase shadowing and indoor coverage problems.

The demand rises from 25 Erlangs at 1,000 subscribers to 2,500 Erlangs at 100,000 subscribers, a 100-fold increase. The selected 74-cell theoretical capacity is about 2,841 Erlangs, so 100,000 subscribers fit only under the ideal load assumptions. A real design should add capacity margin.

Fixed reuse becomes less practical when local traffic is highly uneven because unused channels cannot always be used by a congested cell. Sectoring is useful when interference is the limiting factor. Dynamic channel assignment is useful for temporary or uneven load. Cell splitting becomes preferable when sustained demand in a geographic area exceeds the capacity of its existing cells.



## TASK 6 – Formulate the Recurrence for Repeated Cell Splitting

Question: If overloaded cells are split repeatedly, how can the number of cells, capacity, transmit power and handoff load after k rounds be expressed in terms of round k−1?

**Answer**

In a four-way split, one parent cell of radius R is replaced by four child cells of radius R/2. The area of each child is one quarter of the parent area, so four children cover approximately the same area.

Uniform splitting recurrence: let $M_0$ be the initial number of cells. Then $M_k = 4M_{(k-1)}$, with base case $M_0$. Therefore $M_k = M_0 \times 4^k$.

If each small cell can support the same traffic capacity $C_0$ as a cell with the same channel resources, the capacity recurrence is $C_k = 4C_{(k-1)}$, so $C_k = C_0 \times 4^k$. In practice, spectrum reuse and interference constraints determine whether the full fourfold increase is achieved.

The assignment gives the transmit-power relation $P' = P/2^n$ for a radius reduction by half. Thus $P_k = P_{(k-1)}/2^n$ and $P_k = P_0/2^{(nk)}$. For n=4, each split round gives P/16; for n=2.5, it gives P/5.657.

Handoff load increases as cells become smaller. A simple planning approximation is $H_k \propto 1/R_k$. Since $R_k=R_0/2^k$, handoff rate per moving user may increase approximately by $2^k$.

Selective splitting: if only a fraction f of the $M_{(k-1)}$ cells is overloaded, then $M_k = M_{(k-1)} + 3fM_{(k-1)} = M_{(k-1)}(1+3f)$. The additional three cells arise because each selected parent is replaced by four children. If f varies by round, $M_k = M_0 \times$ product from r=1 to k of $(1+3f_r)$.

Base case: k=0 is the original layout, with $M_0$ cells, radius $R_0$, power $P_0$ and capacity $C_0$.

## TASK 7 – Solve the Recurrence and Interpret the Result

Question: How many splitting rounds are needed to carry the forecast demand, and what do the closed-form results say about cost and mobility?

**Answer**

For uniform four-way splitting, after k rounds the capacity is $C_k = C_0 \times 4^k$. To support demand D, require $C_0 \times 4^k \geq D$.

Dividing by $C_0$: $4^k \geq D/C_0$. Taking logarithm base 4 gives $k \geq \log_4(D/C_0)$. Therefore the minimum integer number of rounds is $k = \text{ceil}[\log_4(D/C_0)]$.

Example: assume one original cell supports $C_0 = 38.39$ Erlangs and the local demand grows to D = 150 Erlangs. $D/C_0 = 3.91$, so $k = \text{ceil}(\log_4 3.91) = 1$ round. The ideal capacity becomes $4 \times 38.39 = 153.56$ Erlangs.

For selective splitting with a constant overloaded fraction f, capacity multiplier per round is approximately (1+3f). The number of rounds is $k = \text{ceil}[\log(D/C_0)/\log(1+3f)]$. If f=0.5, the multiplier is 2.5 per round.

After k uniform rounds, sites grow by $4^k$, radius falls to $R_0/2^k$, transmit power follows $P_0/2^{(nk)}$, and mobility-related handoff load can rise approximately by $2^k$. The mathematical result shows that capacity can grow quickly, but site cost, backhaul, interference coordination and handoff overhead also increase.

## TASK 8 – Implement and Test the Planning Strategies

Question: Can a simulator built from your own strategies reproduce the behaviour predicted by your mathematics?

**Answer and Python reference implementation**

The following compact Python program calculates Erlang-B blocking, channels per cell, approximate SIR and cell count for the requested subscriber loads. It is a reproducible analytical planning simulator. It does not claim to be a full geographic Monte-Carlo simulator with measured antenna patterns or real terrain.

```
import math
S = 336
area = 12
```

```
R = 0.25

subscribers = [1000, 10000, 20000, 100000]

cluster_sizes = [3, 4, 7, 12]

path_loss = [2.5, 3, 4]


def erlang_b(channels, traffic):
    b = 1.0
    for m in range(1, channels + 1):
        b = traffic * b / (m + traffic * b)
    return b


def max_traffic(channels, target=0.02):
    low, high = 0.0, channels * 3.0
    for _ in range(60):
        mid = (low + high) / 2
        if erlang_b(channels, mid) > target:
            high = mid
        else:
            low = mid
    return low


cell_area = 2.598 * R * R

cells = math.ceil(area / cell_area)


print("Cells required:", cells)
```

```
for n in cluster_sizes:
    channels = S // n
    capacity = max_traffic(channels)
    q = math.sqrt(3 * n)
    print("\nN =", n, "channels/cell =", channels,
          "traffic/cell =", round(capacity, 2))
    for exponent in path_loss:
        sir = (3 * n) ** (exponent / 2) / 6
        sir_db = 10 * math.log10(sir)
        print("n =", exponent, "SIR =", round(sir_db, 2), "dB")
    for users in subscribers:
        offered = users * 0.025
        total_capacity = cells * capacity
        print(users, "users:", round(offered, 2), "Erlangs;",
              "capacity:", round(total_capacity, 2),
              "blocking target met:", offered <= total_capacity)
```

Test cases: N = 3, 4, 7 and 12; path-loss exponent n = 2.5, 3 and 4; and subscriber levels 1,000, 10,000, 20,000 and 100,000. The same formulas and input values are used for each comparison.

Correctness check: for N=7 and n=4, the six-interferer approximation gives SIR = 18.66 dB, which is approximately the 18.7 dB value stated in the assignment.

To extend this into a full Monte-Carlo simulator, generate random mobile positions in hexagonal cells, calculate distance-based received power from the serving and co-channel base stations, include log-normal

shadowing, and repeat for many random positions. Compare the mean and 5th-percentile SIR with the analytical estimate.

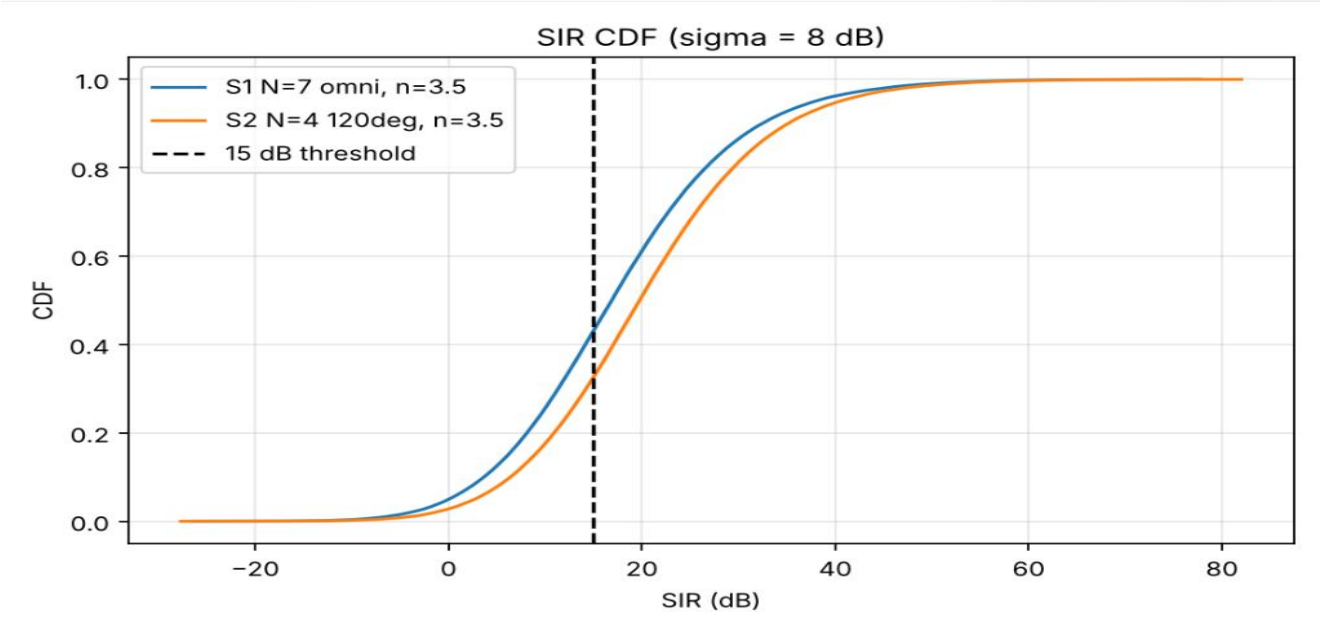
## TASK 9 – Record and Present Experimental Performance

Question: Which measurements best expose the differences between your strategies, and how should they be presented?

**Answer**

| **Measurement** | **What it shows** | **Recommended presentation** |
|---|---|---|
| Mean SIR and 5th-percentile SIR | Typical and cell-edge interference performance | SIR summary table and CDF plot |
| Outage probability | Fraction of users below the SIR threshold | Bar chart by strategy |
| Blocking probability | Probability a call cannot obtain a channel | Blocking vs offered-load plot |
| Channels per cell | Spectrum available to each cell/sector | Comparison table |
| Number of sites | Coverage and deployment scale | Bar chart or table |
| Runtime | Computational cost of the simulator | Runtime comparison table |

For a fair comparison, use the same subscriber locations, random seeds, propagation assumptions and traffic levels for all strategies. Report N, n, sectoring configuration and channel count beside each result. The analytical Erlang-B reference for 48 channels is approximately 38.39 Erlangs at 2% blocking.




## TASK 10 – Compare Theory with Simulation

Question: Where do the simulated results agree with analytical predictions, where do they differ, and what analyses the gap?

**Answer**

The analytical model predicts the broad trend: increasing N increases reuse distance and SIR, but reduces channels per cell. Reducing R increases the number of cells approximately as $1/R^2$. Erlang-B predicts blocking from the offered traffic and number of channels.

A detailed simulation may differ because:

- The six-interferer model assumes equal distances and equal interference powers; real interferers are at different distances.
- Shadowing and fading cause received power to vary around the path-loss prediction.
- A finite network has fewer surrounding interferers at its outer boundary than an infinite hexagonal grid.

- Mobile positions are not always uniformly distributed; people cluster in lecture halls, hostels and event venues.
- Erlang-B assumes a particular loss-system traffic model with Poisson arrivals and no waiting queue; real call durations and handoffs may differ.
- Cluster size and channel counts must be integers, so rounding changes the exact allocation.

The strategy expected to scale best is dynamic channel assignment combined with selective cell splitting. Fixed reuse is easier to operate but cannot respond well to local surges. Sectoring improves interference performance. Dynamic allocation handles short-term imbalance, while splitting adds long-term capacity. The final choice should be confirmed with the same-seed simulation described in Task 8.

## TASK 11 – Recommend and Defend the Final Design

Question: Which design would you actually deploy in the township, and what evidence makes it better than the alternatives?

**Answer**

Recommended design: use a seven-cell frequency reuse pattern (N=7) as the baseline, with a nominal radius of 0.25 km and approximately 74 cells across 12 km². Use sector antennas in open or interference-limited areas, dynamic channel assignment for local traffic variation, and selective cell splitting in permanently overloaded locations.

| **Design item** | **Recommendation** |
|---|---|
| Cluster size | N = 7 baseline |
| Reuse ratio | $Q = \sqrt{21} = 4.583$ |
| Reuse distance at R=0.25 km | Approximately 1.146 km |
| Channels per cell | 48 average |

| | |
|---|---|
| Cell area | Approximately 0.1624 km² |
| Estimated cells | Approximately 74 |
| Per-cell offered traffic at 2% blocking | Approximately 38.39 Erlangs for 48 channels |
| SIR reference | 18.66 dB at n=4 with six equal first-tier interferers; sectoring or larger N needed in open n=2.5 areas |

Why this design? N=7 provides a useful balance between channel availability and interference. Sectoring improves SIR in open areas. Dynamic channel assignment improves use of idle channels. Selective splitting is less expensive than splitting every cell because only overloaded locations receive new sites.

Trade-offs: a larger N improves SIR but gives fewer channels per cell; sectoring improves spatial filtering but adds equipment and can reduce channel pooling; smaller cells improve capacity but increase site cost and handoffs. The design can theoretically support 100,000 subscribers in the ideal 74-cell Erlang-B estimate, but it should not be accepted without a capacity margin and detailed traffic simulation.

## TASK 12 – Day-4 Design Challenge: Re-evaluate and Adapt the Design

Question: A convocation/sports event brings 25,000 attendees, each generating 0.05 Erlang, into a 1 km² stadium zone for four hours. Event days occur roughly twice a month. Should the original design be retained, modified or redesigned?

**A. Impact Analysis**

Event traffic = 25,000 × 0.05 = 1,250 Erlangs.

Event traffic density = 1,250 Erlangs / 1 km² = 1,250 Erlangs/km².

For comparison, 100,000 subscribers spread uniformly across 12 km² generate 100,000 × 0.025 = 2,500 Erlangs, or approximately 208.3 Erlangs/km². The stadium event density is about 1,250/208.3 = 6 times higher.

The original uniform-traffic and steady-demand assumptions no longer hold in the stadium. Demand is concentrated in a small area and lasts four hours. Because the event is predictable and temporary, a temporary capacity solution is more economical than permanently redesigning the entire township.

**B. Evaluation of the Original Design**

The original 1 km² stadium area contains approximately 1/0.1624 = 6.16, or about 7 cell-equivalents at R=0.25 km. At 48 channels per cell and 2% blocking, its ideal offered capacity is about 7 × 38.39 = 268.7 Erlangs. The event offers 1,250 Erlangs, which is about 4.65 times this ideal capacity.

Therefore, the unchanged design cannot meet the 2% blocking target in the stadium. Blocking would become unacceptable because the offered traffic is far above the available channel capacity. Interference may also increase if neighbouring cells attempt to reuse channels too closely.

**C. Adaptation or Redesign**

Decision: MODIFY the original design temporarily during event periods, rather than permanently redesigning the whole township.

- Deploy temporary microcells or cells-on-wheels (COWs) around the stadium.
- Use sector antennas, for example three 120-degree sectors, to focus capacity into the crowd area.
- Use dynamic channel borrowing from lightly loaded nearby cells, only after an interference check.
- Apply temporary power and antenna optimisation; coordinate frequencies with surrounding macro cells.
- Provide sufficient backhaul and power for the four-hour event, and remove or deactivate temporary sites after the event.

Capacity estimate: the event requires 1,250 Erlangs. At 48 channels per cell, the approximate 2%-blocking capacity is 38.39 Erlangs per cell. Required cell-equivalents = 1,250/38.39 = 32.56, so at least 33 equivalent 48-channel cell resources are required under the simplified Erlang-B model. The existing stadium coverage provides about 7 cell-equivalents, so roughly 26 additional equivalent cell resources are

needed. This is a planning estimate; sector resource pooling, channel borrowing and actual traffic distribution can change the required number.

A temporary COW/microcell solution is preferred because the event occurs only about twice a month. Permanent extra sites would create unnecessary operating and maintenance costs during normal days.

**D. Evidence-Based Defence**

The event creates 1,250 Erlangs in 1 km², six times the comparison township-wide average traffic density. The original local capacity estimate of about 269 Erlangs is far below the event load. Therefore retaining the original design unchanged is not suitable.

A temporary modification is the best balance: it adds capacity only when needed, improves coverage and SIR through localised cells and sectoring, and avoids permanent costs. The operator must validate the final number of temporary cells using a site survey, channel plan, backhaul check and event-day simulation. The event lasts four hours and occurs twice monthly, so temporary capacity is more economical than permanent stadium-wide cell splitting.

**Overall Conclusion**

A hexagonal cellular network can share a fixed spectrum through frequency reuse, sectorisation and dynamic channel assignment. The cluster size controls the balance between capacity and interference. With N=7 and R=0.25 km, the ideal planning estimate is about 74 cells and 48 channels per cell. Erlang-B gives approximately 38.39 Erlangs per cell at 2% blocking, or about 2,841 Erlangs for the full idealised layout.

The recommended township design combines N=7 fixed reuse, sectoring where required, dynamic channel assignment and selective cell splitting. For the stadium event, temporary microcells/COWs and channel borrowing should be used because the event traffic is concentrated and short-lived. Real

deployment requires measured propagation data, site costs, detailed simulation and operational capacity margins.